

A Machine-Verified Proof of TLMC Conjecture #590: Theta Series of Numerical Semigroups

Batch 11 solution submission

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Abstract

We *prove* a conjecture of the Last Math Competition: for a numerical semigroup S with Frobenius number F , the theta series $\Theta_S(q) = \sum_{s \in S} q^s$ satisfies $\Theta_S(q) - \frac{1}{1-q} = -\sum_{g \in \text{Gaps}} q^g$, a polynomial with at most F monomials (the gaps all lie in $\{1, \dots, F\}$ since $0 \in S$ and $n > F \Rightarrow n \in S$). The Lean 4 formalization defines the gap polynomial as an explicit monomial sum, realizes $1/(1-q)$ via the geometric power series (`ones`, with the kernel-verified identity $(1-q) \cdot \text{ones} = 1$), proves the coefficient identity of $\Theta_S + G = 1/(1-q)$, and bounds the support. Compilation is clean; standard axioms only; no `sorry`.

1 The conjecture

Conjecture 1.1 (TLMC-00000000590). For a numerical semigroup S with Frobenius number F : $\Theta_S(q) - \frac{1}{1-q}$ is a polynomial with at most Frobenius-many monomials (a sparsity bound).

2 The proof

Theorem 2.1. *Conjecture 1.1 is **true**. More precisely, for $S \subseteq \mathbb{N}$ with $0 \in S$, $F \notin S$ and the conductor property $n > F \Rightarrow n \in S$:*

$$\Theta_S(q) + G(q) = \frac{1}{1-q}, \quad G(q) := \sum_{\substack{g \leq F \\ g \notin S}} q^g,$$

and G has at most F monomials.

Proof. The identity is coefficientwise: at n , the n -th coefficient of $\frac{1}{1-q}$ is $1 = [n \in S] + [n \leq F \wedge n \notin S]$ (if $n \in S$ the first indicator is 1 and the second is 0; if $n \notin S$ then $n \leq F$ by the conductor property, and the second indicator is 1). For the sparsity: $\text{Supp}(G) = \{g \leq F : g \notin S\}$ has at most F elements, since it is contained in $\{1, \dots, F\}$ ($0 \in S$ excludes 0) — formally, the gap set is a proper subset of $\{0, \dots, F\}$ by the exclusion of 0, hence has at most F elements. $\square$

3 Machine verification

`Batch11.lean` (namespace `TLMCBatch11`) contains: `ones` and `one_sub_mul_ones` (the geometric series with $(1-q) \cdot \text{ones} = 1$, proved via antidiagonal coefficient sums), `thetaS`, `gapPoly` with

`coeff_gapPoly` (its coefficients are the gap indicators, via `sum_eq_single_of_mem`), and the main theorem `tlm590` packaging the identity $\Theta_S + \iota(G) = \text{ones}$ and the bound $\#\text{Supp}(G) \leq F$. Compilation via `lake env lean Batch11.lean` exits 0 with zero errors and warnings; `#print axioms` reports at most `[propext, Classical.choice, Quot.sound]`; no sorry.